

Stable Exponent Gain for Schur Multipliers Toward Coine’s Endpoint Problem

Automated mathematics run `fa_banach_001`, lane 7

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Status. Substantial partial result, likely valid, pending human review. The result proves a new exponent-gain inclusion and gives a direct proof of the known $p = 2$ endpoint. It does not settle the question for $1 < p < 2$.

1 Source question and result

Coine’s Problem 5.9 asks whether

$$\mathcal{M}(p, 1) = \mathcal{M}(p, p), \quad 1 < p \leq 2,$$

for Schur multipliers on the indicated sequence-space operator classes [1]. The exact question appears on PDF page 24:

concludes the proof.

□

Problem 5.9. Compare the other spaces of Schur multipliers. For example, if $1 < p \leq 2$, do we have

$$\mathcal{M}(p, 1) = \mathcal{M}(p, p)?$$

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Figure 1: Problem 5.9 in [1].

The equality at $p = 2$ follows from Bennett’s inclusion theorem as quoted by Coine. For $1 < p < 2$, Bennett’s equality region reaches every $\mathcal{M}(p, q)$ with $q < p$ but not the diagonal endpoint $q = p$. The following result places a substantial family of diagonal multipliers on the desired $\mathcal{M}(p, 1)$ side.

For an operator $T : E \rightarrow F$, write $\gamma_r(T)$ for its L^r -factorization norm. Write $L_{1,p'}(T)$ for the infimum of $\|S\|\|R\|$ over factorizations

$$E \xrightarrow{R} L^p(\Omega) \xrightarrow{I_{p,1}} L^1(\Omega) \xrightarrow{S} F$$

on probability spaces. Coine’s Theorem 4.2 and Corollary 4.5 identify these operator norms with the relevant Schur multiplier norms.

Theorem 1 (Stable exponent gain). *Let $1 < p < r \leq 2$. Every operator $T : \ell_1 \rightarrow \ell_\infty$ which factors through an L^r space factors through a probability-space inclusion $L^p \hookrightarrow L^1$. Quantitatively,*

$$L_{1,p'}(T) \leq \frac{\kappa_{r,p}}{\kappa_{r,1}} \gamma_r(T),$$

where $\kappa_{r,s} = (\mathbb{E}|\theta_r|^s)^{1/s}$ and θ_r is a normalized symmetric r -stable scalar. Consequently,

$$\boxed{\mathcal{M}(r, r) \subseteq \mathcal{M}(p, 1)}.$$

In particular, $\mathcal{M}(2, 2) \subseteq \mathcal{M}(p, 1)$ for every $1 < p < 2$. At $p = r = 2$, with Gaussian constants in place of $\kappa_{r,s}$,

$$\mathcal{M}(2, 1) = \mathcal{M}(2, 2).$$

2 Proof intuition

An r -stable random measure turns every vector in an L^r factor into a random variable whose L^s norm is an exact constant times the original L^r norm for every $s < r$. Taking $s = p$ makes the new input map bounded into L^p ; taking $s = 1$ makes the same map bounded below in L^1 . Hence the old output map is bounded on the resulting subspace when that subspace is given its L^1 norm. The special range ℓ_∞ now matters: coordinatewise Hahn–Banach extends this output map to all of L^1 without increasing its norm. This produces exactly the factorization characterizing $\mathcal{M}(p, 1)$.

The strict inequality $p < r$ is structural. A nondegenerate r -stable variable has finite s -moment precisely for $s < r$ (apart from the Gaussian case $r = 2$), so the same mechanism has no L^p endpoint when $r = p < 2$.

3 Proof of Theorem 1

Fix a factorization

$$T = BA, \quad \ell_1 \xrightarrow{A} L^r(\Sigma, \nu) \xrightarrow{B} \ell_\infty.$$

Only the closed linear span

$$X = \overline{A(\ell_1)} \subseteq L^r(\Sigma, \nu)$$

is relevant, and X is separable.

For $1 < r < 2$, the standard symmetric r -stable random-measure construction gives a probability space $(\Omega, \mathbb{P})$ and a complex-linear map $J : X \rightarrow L^0(\Omega)$ such that for every $x \in X$, Jx is an isotropic r -stable scalar of scale $\|x\|_r$. Thus, for every $1 \leq s < r$,

$$\|Jx\|_{L^s} = \kappa_{r,s} \|x\|_r. \tag{1}$$

This is the usual stable-process embedding of L^r ; see [3, Chapter 3] and the stable-process discussion used in [2, Theorem 6.4.19]. For $r = 2$, take the isonormal complex Gaussian process over the Hilbert space X ; then (1) holds for every finite s , with Gaussian moment constants.

Since $p < r$, equation (1) shows that

$$J : X \longrightarrow L^p(\Omega) \quad \text{has norm } \kappa_{r,p},$$

while the same formula with $s = 1$ shows that JX is a closed subspace of $L^1(\Omega)$ and that $J^{-1} : JX \rightarrow X$ has norm $\kappa_{r,1}^{-1}$ when JX carries its inherited L^1 norm.

Define

$$S_0 : JX \longrightarrow \ell_\infty, \quad S_0(Jx) = Bx.$$

This is well-defined and

$$\|S_0\| \leq \frac{\|B\|}{\kappa_{r,1}}.$$

The space ℓ_∞ is 1-injective: writing the coordinate functionals of S_0 and extending each by Hahn–Banach gives an operator

$$S : L^1(\Omega) \longrightarrow \ell_\infty \quad \text{with} \quad \|S\| = \|S_0\|.$$

Set $R = JA : \ell_1 \rightarrow L^p(\Omega)$. Because the middle map is the canonical probability-space inclusion $I_{p,1} : L^p \hookrightarrow L^1$, we have

$$T = SI_{p,1}R.$$

Moreover,

$$\|S\|\|R\| \leq \frac{\kappa_{r,p}}{\kappa_{r,1}} \|B\|\|A\|.$$

Taking the infimum over all L^r factorizations proves the operator-norm inequality.

For a matrix symbol ϕ , Coine’s characterization gives

$$\|\phi\|_{\mathcal{M}(r,r)} = \gamma_r(u_\phi), \quad \|\phi\|_{\mathcal{M}(p,1)} = L_{1,p'}(u_\phi).$$

The claimed Schur-multiplier inclusion and estimate follow.

When $p = r = 2$, use the Gaussian construction, for which both the first and second moments are finite, to obtain $\mathcal{M}(2,2) \subseteq \mathcal{M}(2,1)$. Conversely, a factorization through $L^2 \hookrightarrow L^1$ is in particular a factorization through L^2 after composing the last two arrows. Hence $\mathcal{M}(2,1) \subseteq \mathcal{M}(2,2)$, completing the endpoint proof. $\square$

4 Verification, limitations, and upgrade audit

The proof is analytic and has no computational dependency. Its two nonformal inputs are standard: the stable random-measure embedding and the 1-injectivity of ℓ_∞ . All remaining arrows and constants are displayed explicitly.

This theorem does not prove $\mathcal{M}(p,p) \subseteq \mathcal{M}(p,1)$ for $1 < p < 2$ and does not give a counterexample. The exponent gap cannot simply be closed: a nondegenerate p -stable variable has infinite p -th moment. The associated universal-symbol reduction also stops short of a contradiction. If $\sigma : \ell_1 \twoheadrightarrow \ell_p$ is a metric quotient and $V : \ell_p \hookrightarrow \ell_\infty$ is an isometry, then $V\sigma \in \Gamma_p$; a hypothetical $(p,1)$ factorization yields only a factorization of I_{ℓ_p} through a subspace of a quotient attached to $L^p \hookrightarrow L^1$. It does not yield a direct factorization through that inclusion.

Eight focused upgrade attempts were made. Besides the successful stable gain, they tested: the Hilbert endpoint; the universal identity symbol; Reisner’s $(p,1)$ Gordon–Lewis obstruction; finite-dimensional quotient/isometry amplification; Hadamard blocks; the source’s Rademacher type estimate at its zero-exponent endpoint; and truncation of p -stable variables. The detailed log records why none of the remaining routes closes the question.

5 Literature and novelty boundary

Bounded searches on 17 August 2026 covered the run indexes, the exact arXiv id/title/Problem 5.9 wording, Coine’s published paper and thesis, Bennett’s inclusion cases, and combinations of “ Γ_r ”, “ $(p, 1)$ -factorable”, and stable-process factorization. No explicit later resolution of Problem 5.9 and no explicit statement of Theorem 1 was located. Novelty confidence is moderate rather than high because the argument is short and may be implicit in standard operator-ideal folklore.

Human review is recommended, focusing on the simultaneous complex stable embedding, the coordinatewise extension into ℓ_∞ , and the exact translation through Coine’s factorization norms.

References

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- [4] S. Reisner, *Operators factoring through Banach lattices and ideal norms*, Serdica Math. J. 21 (1995), 75–89.